

The Sharp Universal Euler Constant for Harmonic Polylogarithms

Positive divided differences, exact certificates,
and all-order error-moment identities

Research continuation for the ProveIt project

10 October 2026

Abstract

For $a, b > 0$ let $g_{a,b} = \operatorname{Im} \operatorname{Li}_{a,b}(i, 1)$, interpreted by principal analytic continuation, and let $E_N(a, b)$ be the N -term Euler transform of its odd harmonic coefficient sequence. We prove the parameter-free bound

$$0 < E_N(a, b) - g_{a,b} < \frac{9}{8} 2^{-N} \quad (N \geq 2).$$

The proof reduces the entire parameter range to a two-variable rational kernel. Two finite polynomial positivity certificates and one elementary envelope argument treat every truncation length; the certificates comprise 1542 strictly positive rational Bernstein coefficients. Consequently the sharp constant uniform in $a, b > 0$ and $N \geq 1$ is

$$C_* = \max_{b>0} \{\beta(b) + 2^{-b}\eta(b)\},$$

where β and η are the Dirichlet beta and eta functions. This identifies the previously unresolved global Euler constant with the manuscript's already established Gaussian axis maximum. Exact arithmetic additionally proves $1.3021 < b_* < 1.3023$ and $1.1365611033 < C_* < 1.1365611046$. We derive a generating function for all Euler errors and exact binomial-moment identities that lower the outer polylogarithm order. They give new explicit convergent identities and extend Gaussian negativity to every real outer order at least -1 . The report distinguishes inherited results, new proofs, exact computation and numerical diagnostics, and supplies proposed manuscript edits and independent verifiers. The mixed-period S_6 conjecture is not resolved here.

Source snapshot. The analysis is relative to ProveIt commit `3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f`, the reviewed real-order manuscript milestone of 10 October 2026. This is an AI-assisted research contribution, not an externally refereed paper or a proof-assistant formalization. No assertion of literature priority is made.

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1 The problem selected from the manuscript

1.1 Notation and the boundary-value convention

We use the decreasing-index convention

$$F_{a,b}(z) = \text{Li}_{a,b}(z, 1) = \sum_{n=1}^{\infty} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad H_0^{(b)} = 0. \quad (1.1)$$

Initially $|z| < 1$ and $a, b > 0$. The positive integral below continues F to $\mathbb{C} \setminus [1, \infty)$. In particular,

$$g_{a,b} = \text{Im } F_{a,b}(i)$$

always denotes this analytic value. The ordinary Fourier series at i need not converge when the coefficients in (1.1) fail to tend to zero. Nothing in the following proofs requires it to converge.

Set

$$A_k(a, b) = \frac{H_{2k}^{(b)}}{(2k+1)^a}, \quad d_j(a, b) = \sum_{k=0}^j (-1)^k \binom{j}{k} A_k(a, b), \quad E_N(a, b) = \sum_{j=0}^{N-1} \frac{d_j(a, b)}{2^{j+1}}. \quad (1.2)$$

Thus $A_0 = d_0 = E_1 = 0$. We also use $E_0 = 0$ and distinguish the unscaled and scaled errors

$$e_N = E_N - g_{a,b}, \quad R_N = 2^N e_N \quad (N \geq 0). \quad (1.3)$$

That distinction matters: e_N decreases, but R_N need not do so.

The auxiliary parameter b is a real order, not a depth. The function has depth two throughout. We retain the standard notation

$$\beta(s) = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^s}, \quad \eta(s) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n^s}$$

when the series converge, and use their analytic continuations otherwise. The elementary polylogarithm and gamma identities used here are consistent with the principal conventions in the DLMF [6, 7].

1.2 What was already proved, and what remained open

At the pinned snapshot, the manuscript's signed-kernel and real-order chapters establish one-sided Euler convergence, a constant-one estimate for $a \geq 1$, and several parameter-dependent bounds for $0 < a < 1$. The subcritical chapter proves the sharp constant

$$C_{\text{crit}} = \frac{\pi}{4} + \frac{\log 2}{2} \quad (1.4)$$

uniformly on $a, b > 0$, $a + b \leq 1$. It also proves that

$$C(b) = \beta(b) + 2^{-b} \eta(b) \quad (1.5)$$

has a unique nondegenerate global maximum, at an order $b_* \in (1, 2)$, and that this is the supremum of $-2g_{a,b}$ over all positive orders. Those results are inherited, not discoveries of this continuation [1, 2].

The research programme nevertheless explicitly leaves a universal real-order Euler estimate above the critical triangle open. The warning is correct: bounding the Gaussian value only

controls $N = 1$, since $2(E_1 - g) = -2g$. It does not control later scaled errors without another theorem. This is precisely the gap treated here [3].

The new contribution is the uniform estimate for *every* $N \geq 2$. It makes the inherited axis maximum decisive for the full Euler problem. The sharp constant is not guessed by maximizing a finite numerical table. An analytic reduction and exact polynomial certificates exclude all later truncations from competing with the first one.

1.3 Main result and its logical components

Theorem 1.1 (Sharp unrestricted real-order Euler bound). *For all $a, b > 0$ and integers $N \geq 1$,*

$$0 < E_N(a, b) - g_{a,b} < C_* 2^{-N}, \quad C_* = \max_{b>0} C(b). \quad (1.6)$$

The constant C_ is the least constant valid uniformly on this domain. More strongly,*

$$0 < E_N(a, b) - g_{a,b} < \frac{9}{8} 2^{-N} \quad (N \geq 2). \quad (1.7)$$

The sharp supremum in (1.6) is approached with $N = 1$, $b = b_$ and $a \downarrow 0$. If the axis $a = 0$, $b > 0$ is included, the non-strict sharp estimate has equality only at $(a, b, N) = (0, b_*, 1)$.*

The proof of (1.7) occupies Sections 2–3; Section 4 identifies the optimum. Section 5 gives exact rational enclosures, not just numerical approximations. Section 6 derives an independent family of all-order identities from the error generating function.

The underlying Euler transform is classical. Positive-measure arguments for sequence acceleration go back well beyond this project; the framework of Cohen–Rodriguez Villegas–Zagier is especially relevant [4]. Euler-type transformations involving harmonic numbers also have a substantial literature [5]. The present claim is a specific uniform theorem for the harmonic depth-two family, not the invention of Euler acceleration or of positive-kernel methods.

2 A positive divided-difference representation

2.1 Independent gamma variables

For $a, b > 0$, let S and T be independent gamma variables of shapes a, b and rate one, and put

$$X = e^{-S}, \quad Y = e^{-T}, \quad p = X^2.$$

Then $0 < X, Y < 1$ almost surely. The elementary gamma integral gives

$$\mathbb{E}X^{n-1} = n^{-a}, \quad \mathbb{E}Y^{m-1} = m^{-b}. \quad (2.1)$$

Consequently

$$A_k = \mathbb{E} \left[\frac{p^k - (pY^2)^k}{1 - Y} \right]. \quad (2.2)$$

For each fixed integer k , the quotient extends continuously at $Y = 1$: it is $p^k(1 + Y + \dots + Y^{2k-1})$, with the value zero when $k = 0$. Thus its expectation is finite even near $T = 0$.

Summing (1.1) absolutely in the disk yields

$$F_{a,b}(z) = z^2 \mathbb{E} \left[\frac{X}{(1 - zX)(1 - zXY)} \right]. \quad (2.3)$$

For each compact subset of the slit plane, $|1 - zv|$ has a positive lower bound uniform in $0 \leq v \leq 1$. The integrand is then bounded and holomorphic. Dominated differentiation gives the asserted continuation. This argument has no signed-measure threshold.

At the axis $a = 0$, take $S = 0$ and $X = 1$ deterministically. Equations (2.2) and (2.3) remain valid, and

$$F_{0,b}(z) = \frac{z \operatorname{Li}_b(z)}{1 - z}. \quad (2.4)$$

The gamma variables converge to this axis model as $a \downarrow 0$. This probabilistic convention is only a convenient notation for the corresponding positive integrals.

2.2 The remainder kernel

For $N \geq 0$ define

$$f_N(v) = \frac{(1 - v)^N}{1 + v} \quad (0 \leq v \leq 1), \quad K_N(p, y) = \frac{f_N(py^2) - f_N(p)}{1 - y} \quad (0 \leq y < 1). \quad (2.5)$$

For $N \geq 1$, f_N is strictly decreasing on $[0, 1]$. The continuous boundary value at $y = 1$ is

$$K_N(p, 1) = -2pf'_N(p) = \frac{2p(1 - p)^{N-1}\{N + 1 + (N - 1)p\}}{(1 + p)^2}. \quad (2.6)$$

All formulas below use the continuous endpoint interpretation. In particular, no uncontrolled division by a small $1 - y$ is needed in a numerical implementation.

Proposition 2.1 (Exact positive remainder). *For $a \geq 0$, $b > 0$ and $N \geq 1$,*

$$R_N(a, b) = \mathbb{E}K_N(X^2, Y) > 0. \quad (2.7)$$

Furthermore $d_j < 0$ for $j \geq 1$, and the E_N decrease strictly.

Proof. The binomial theorem applied to the finite expression (2.2) gives

$$d_j = \mathbb{E} \left[\frac{(1 - p)^j - (1 - pY^2)^j}{1 - Y} \right].$$

It is strictly negative for $j \geq 1$. For fixed v , the finite geometric identity is

$$\sum_{j=0}^{N-1} \frac{(1 - v)^j}{2^{j+1}} = \frac{1}{1 + v} - 2^{-N} \frac{(1 - v)^N}{1 + v}.$$

Apply it to the difference of the two kernels. On the other hand, rationalizing (2.3) at $z = i$ gives

$$g_{a,b} = \mathbb{E} \left[\frac{(1 + p)^{-1} - (1 + pY^2)^{-1}}{1 - Y} \right] = -\mathbb{E} \left[\frac{p(1 + Y)}{(1 + p)(1 + pY^2)} \right]. \quad (2.8)$$

Subtracting proves (2.7) directly. Every operation so far involves bounded rational functions or finite sums. In particular, an infinite alternating series has not been interchanged with an integral. Strict positivity follows because $p > 0$ and $0 < Y < 1$ almost surely, including on the axis. $\square$

The order variables now occur only in the probability distribution of (p, Y) . A pointwise bound on K_N simultaneously covers supercritical, critical and subcritical orders. This removes the blow-up of the individual Jordan-mass estimates near $a + b = 1$.

Remark 2.2 (What this representation does not say). The signed-moment representation of the original coefficients need not exist below the threshold. The present representation is a positive expectation of a *difference quotient*, not a replacement signed measure with incorrectly assumed finite mass. The cancellation in the numerator is retained throughout.

3 A uniform rational kernel inequality

Theorem 3.1 (Kernel bound for every later truncation). *For every integer $N \geq 2$ and $0 \leq p, y \leq 1$,*

$$0 \leq K_N(p, y) < \frac{9}{8}. \quad (3.1)$$

The lower bound is strict when $p > 0$ and $y < 1$. The strict upper bound includes the continuous boundary values.

The proof has two finite base cases and an all- N envelope. We give enough information to reconstruct every rational certificate without a symbolic algebra system.

3.1 Two explicit polynomial base cases

Multiplication by the positive denominator shows that $K_N < 9/8$ is equivalent to positivity of

$$P_N(p, y) = 8(1+p)(1+py^2)\{9/8 - K_N(p, y)\}.$$

For $N = 2, 3$ these are the following polynomials:

$$\begin{aligned} P_2 = & 8y^2(1+y)p^3 + (8y^3 + 17y^2 + 8y + 8)p^2 \\ & + (9y^2 - 24y - 15)p + 9, \end{aligned} \quad (3.2)$$

$$\begin{aligned} P_3 = & -8y^2(y+1)(y^2+1)p^4 \\ & + (-8y^5 - 8y^4 + 16y^3 + 16y^2 - 8y - 8)p^3 \\ & + (24y^3 + 33y^2 + 24y + 24)p^2 + (9y^2 - 32y - 23)p + 9. \end{aligned} \quad (3.3)$$

An identity check can avoid rational-function division altogether:

$$\begin{aligned} (1-y)P_N = & 9(1-y)(1+p)(1+py^2) \\ & - 8\{(1-py^2)^N(1+p) - (1-p)^N(1+py^2)\}. \end{aligned} \quad (3.4)$$

Both verifiers check this identity or independently reconstruct its quotient.

For a polynomial $P(x, y) = \sum c_{kl}x^k y^l$ of bidegree at most (r, s) , its tensor Bernstein coefficients on the unit square are

$$B_{ij} = \sum_{k \leq i, l \leq j} c_{kl} \binom{i}{k} \binom{j}{l} \binom{r}{k} \binom{s}{l}. \quad (3.5)$$

The Bernstein basis functions are nonnegative and sum to one. Hence a strictly positive minimum of these coefficients proves strict positivity of P on the whole square, including its boundary.

For a mesh of denominator d , first substitute $p = (u + i)/d$, $y = (v + j)/d$ on each cell. Formula (3.5) then gives an exact certificate. The data are summarized by

Polynomial	Bidegree	Mesh	Coefficients	Smallest coefficient
P_2	(3, 3)	8×8	1024	631/12288
P_3	(4, 5)	4×4	480	11337/40960

Every coefficient is stored as a rational number in the accompanying JSON file. More importantly, the standard-library verifier regenerates all of them from (3.2)–(3.5); it does not trust a stored minimum. These finitely many exact additions and multiplications prove the two base cases.

3.2 An envelope for binomial differences

Introduce

$$J_m(p, y) = \frac{(1 - py^2)^m - (1 - p)^m}{1 - y}, \quad m \geq 2.$$

For fixed $0 < y < 1$, differentiation in p gives a unique interior maximum at

$$p_m(y) = \frac{1 - y^{2/(m-1)}}{1 - y^{2m/(m-1)}}. \quad (3.6)$$

Indeed the derivative changes sign exactly when $1 - p = y^{2/(m-1)}(1 - py^2)$. Its sign is positive at $p = 0$ and negative at $p = 1$. Substitution yields the exact envelope

$$J_m(p, y) \leq B_m(y) := (1 + y) \left(\frac{1 - y^2}{1 - y^{2m/(m-1)}} \right)^{m-1}. \quad (3.7)$$

The endpoint values follow by continuity.

Lemma 3.2 (The binomial envelope decreases in its index). *For every $0 < y < 1$, $B_{m+1}(y) < B_m(y)$ for $m \geq 2$.*

Proof. Put $k = m - 1$, $t = -2 \log y > 0$ and $h(v) = \log(1 - e^{-(1+v)t})$. Direct differentiation gives $h''(v) < 0$ for $v \geq 0$. Apart from $\log(1 + y)$, the logarithm of (3.7) is

$$k\{h(0) - h(1/k)\} = -\frac{h(1/k) - h(0)}{1/k}.$$

The secant slope of a strictly concave function decreases as the right endpoint increases. Thus the displayed negative secant slope decreases strictly when k increases. Exponentiation proves the claim. $\square$

3.3 The fourth envelope is below nine eighths

Set $y = t^3$, $0 \leq t \leq 1$. The inequality $B_4(y) < 9/8$ is equivalent away from $t = 1$ to

$$9(1 - t^8)^3 - 8(1 + t^3)(1 - t^6)^3 = (1 - t^2)^3 Q(t) > 0, \quad (3.8)$$

where

$$\begin{aligned} Q(t) = & 9t^{18} + 27t^{16} - 8t^{15} + 54t^{14} - 24t^{13} + 82t^{12} - 48t^{11} \\ & + 84t^{10} - 56t^9 + 60t^8 - 48t^7 + 34t^6 - 24t^5 + 6t^4 - 8t^3 + 3t^2 + 1. \end{aligned} \quad (3.9)$$

The degree-18 Bernstein coefficients after $t = u/2$ and $t = (1 + u)/2$ have respectively the strictly positive minima

$$\frac{192789}{262144}, \quad \frac{690189}{1949696}. \quad (3.10)$$

There are only 38 coefficients. Formula (3.5) with one variable regenerates them. At $y = 1$ the envelope limit is $B_4(1) = 2(3/4)^3 = 27/32 < 9/8$, so the endpoint is covered as well.

3.4 The infinite family follows without an infinite computation

Since $1/(1+v) = \sum_{j \geq 0} (1-v)^j / 2^{j+1}$ on $[0, 1]$,

$$K_N(p, y) = \sum_{j=0}^{\infty} 2^{-j-1} J_{N+j}(p, y). \quad (3.11)$$

For $y < 1$ this follows by subtracting the two convergent geometric series. If $N \geq 4$, (3.7), Lemma 3.2 and (3.8) give

$$J_{N+j}(p, y) \leq B_{N+j}(y) \leq B_4(y) < 9/8.$$

Averaging proves $K_N \leq B_4 < 9/8$ for every $N \geq 4$. Continuity covers $y = 1$. The polynomial base cases cover $N = 2, 3$, completing Theorem 3.1.

Taking expectations in Proposition 2.1 proves (1.7). It also proves $E_N \rightarrow g$ throughout the full parameter domain, with a uniform geometric rate. There is no dependence on the masses of positive and negative parts of a signed measure.

Remark 3.3 (Nature of the finite computation). The inequality is a computer-assisted ordinary proof: three stated polynomial identities, a stated rational basis conversion, and 1542 strictly positive rational numbers close its finite component. The all-index reduction is analytic. The check is reproducible under Python's optimized mode and is not a test of finitely many real sample points. The supplied SymPy replay independently derives the two rational kernels rather than importing their coefficient arrays.

4 Identification of the sharp constant

4.1 Only the first truncation can determine the optimum

The first kernel has the particularly simple form

$$K_1(p, y) = \frac{2p(1+y)}{(1+p)(1+py^2)}. \quad (4.1)$$

Its derivative in p has the sign of $1 - p^2 y^2$. Hence it is strictly increasing for $0 < p < 1$, $0 \leq y < 1$. For $a > 0$, $X^2 < 1$ almost surely, so

$$R_1(a, b) < \mathbb{E}K_1(1, Y) = \mathbb{E} \frac{1+Y}{1+Y^2} = C(b). \quad (4.2)$$

The last equality can also be checked without summing an integral kernel: residue classes give $\text{Li}_b(i) = -2^{-b}\eta(b) + i\beta(b)$, and (2.4) gives $-2g_{0,b} = C(b)$. The two Dirichlet series converge for every $b > 0$.

The gamma variable S tends to zero in probability as $a \downarrow 0$, for example by $\mathbb{E}S = a$ and Markov's inequality. Since K_1 is bounded and continuous, (4.2) tends to equality. Thus the exact supremum at the first truncation is the supremum of the axis function.

For completeness, there is a strict elementary separation between this supremum and the later-truncation bound:

$$C(1) = \frac{\pi}{4} + \frac{\log 2}{2} > \frac{9}{8}. \quad (4.3)$$

One exact verification uses Machin's identity and the positive artanh series:

$$\frac{\pi}{4} = 4 \arctan(1/5) - \arctan(1/239) > 4 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} \right) - \frac{1}{239} > \frac{25}{32},$$

$$\log 2 = 2 \sum_{j \geq 0} \frac{3^{-2j-1}}{2j+1} > 2 \left(\frac{1}{3} + \frac{1}{81} \right) > \frac{11}{16}.$$

Adding proves (4.3). The last two comparisons are checked as rational inequalities by the verifier.

4.2 An ordinary proof of the inherited axis maximum

The existence, uniqueness and nondegeneracy of the maximum were established in the manuscript [1]. We include a short self-contained argument, both to expose the exact dependency and to justify the localization procedure in the next section.

Proposition 4.1 (Unique axis maximum). *There is exactly one stationary point $b_* > 0$ of C . It is a nondegenerate global maximum. Moreover, $C'(b) > 0$ for $b < b_*$ and $C'(b) < 0$ for $b > b_*$.*

Proof. Write

$$C(b) = \frac{1}{\Gamma(b)} \int_0^\infty T^{b-1} e^{-T} \phi(T) dT, \quad \phi(T) = \frac{1 + e^{-T}}{1 + e^{-2T}}.$$

The function ϕ is strictly above one on $(0, \infty)$, tends to one at both endpoints, and increases and then decreases with unique maximum

$$\phi(\log(1 + \sqrt{2})) = \frac{1 + \sqrt{2}}{2}.$$

These assertions follow by differentiating $(1 + y)/(1 + y^2)$ on $0 < y < 1$. Thus $1 < C(b) < (1 + \sqrt{2})/2$ for every $b > 0$.

As $b \downarrow 0$, a gamma variable of shape b tends to zero in probability. As $b \rightarrow \infty$, it tends to infinity in probability: its mean and variance are both b , so Chebyshev's inequality suffices. Bounded convergence in probability yields $C(b) \rightarrow 1$ at both ends. Therefore an interior maximum exists. Differentiation under the integral is valid locally in $b > 0$, since all required logarithmic moments are integrable.

Consider any stationary point b_0 , and put $\lambda = C(b_0)$. The function $f(T) = \phi(T) - \lambda$ has exactly two roots $T_1 < T_2$ and sign pattern $-, +, -$. Define

$$M(b) = \int_0^\infty f(T) T^{b-1} e^{-T} dT = \Gamma(b) \{C(b) - \lambda\}.$$

Then $M(b_0) = M'(b_0) = 0$. Consequently

$$\begin{aligned} M''(b_0) &= \int_0^\infty f(T) T^{b_0-1} e^{-T} (\log T - \log T_1)(\log T - \log T_2) dT \\ &< 0. \end{aligned}$$

The product has strictly negative sign except at the two roots. As $M''(b_0) = \Gamma(b_0)C''(b_0)$, every stationary point is a strict nondegenerate local maximum. Two such maxima would force an intervening local minimum and hence another stationary point that is not a strict maximum. There is therefore exactly one stationary point. Continuity of C' and the endpoint limits give its asserted signs. $\square$

Proof of Theorem 1.1. For $N = 1$, (4.2) gives $R_1(a, b) < C(b) \leq C_*$. For $N \geq 2$, Theorem 3.1 gives $R_N < 9/8 < C(1) \leq C_*$. Positivity is Proposition 2.1. To prove optimality, fix $b = b_*$ and let $a \downarrow 0$ with $N = 1$; the scaled error tends to C_* . Thus no smaller constant works. On the included axis, equality is possible at $N = 1$ only, and Proposition 4.1 makes b_* the unique order where it occurs. $\square$

Remark 4.2 (Sharpness on different domains). The value $9/8$ is a convenient rigorous separator, not claimed to be optimal for $N \geq 2$. The constant one remains the manuscript's sharp uniform constant on its restricted domain $a \geq 1$; the smaller critical constant (1.4) remains sharp on the critical/subcritical triangle. The new theorem answers a different optimization question: simultaneous uniformity over all positive real orders and all truncation lengths.

5 Certified localization and a usable numerical budget

5.1 Exact finite Euler evaluation on the axis

To turn the characterization of C_* into a short rational enclosure, we use only finite Euler sums and the just-proved remainder bound. Rearranging (1.2) gives

$$E_N = 2^{-N} \sum_{k=0}^{N-1} (-1)^k A_k \sum_{j=k+1}^N \binom{N}{j}. \quad (5.1)$$

The finite coefficient identity behind this formula is

$$\sum_{j=k}^{N-1} 2^{N-j-1} \binom{j}{k} = \sum_{j=k+1}^N \binom{N}{j}.$$

It follows by induction from Pascal's identity. The supplied check compares the two implementations for all $1 \leq N \leq 32$, or 528 individual coefficients.

On the axis $a = 0$, $A_k = H_{2k}^{(b)}$ and $C(b) = -2g_{0,b}$. Suppose exact rational arithmetic encloses the finite sum by $L \leq E_N \leq U$. For $N \geq 2$, the kernel theorem implies

$$-2U < C(b) < -2L + \frac{9}{4} 2^{-N}. \quad (5.2)$$

This is an enclosure of an analytic value, even if the original alternating series diverges.

For a positive rational order $b = p/q$ in lowest terms, each m^{-b} is enclosed on a grid of denominator D by an integer r satisfying

$$r^q m^p \leq D^q < (r+1)^q m^p. \quad (5.3)$$

These integer inequalities prove $r/D \leq m^{-b} < (r+1)/D$. All powers, harmonic sums, signs and Euler weights are then combined with rational outward bounds. A decimal computation may propose r when generating a certificate; the verifier accepts it only after checking (5.3) with integers. It never calls a transcendental evaluator.

We use $N = 48$, $D = 10^{30}$, and the three orders 1.3021, 1.3022, 1.3023. There are 94 power brackets per order, hence 282 in total. The following displayed intervals are deliberately rounded outward more coarsely than the stored rational endpoints:

b	Lower bound for $C(b)$	Upper bound for $C(b)$
1.3021	1.1365611027504	1.1365611027505
1.3022	1.1365611033275	1.1365611033277
1.3023	1.1365611030380	1.1365611030381

The middle lower endpoint exceeds both outer upper endpoints. Strict unimodality therefore gives

$$1.3021 < b_* < 1.3023. \quad (5.4)$$

For example, $b_* \leq 1.3021$ would put all three points on the decreasing side and contradict the first comparison. The other exclusion is analogous.

5.2 A curvature bound closes the global enclosure

Lemma 5.1 (A sufficient explicit curvature bound). *For $1 \leq b \leq 2$, $|C''(b)| < 1$.*

Proof. For a gamma variable T of shape b , set $L = \log T - \psi(b)$. Differentiation of its density gives

$$\mathbb{E}L = 0, \quad \mathbb{E}L^2 = \psi_1(b), \quad C''(b) = \mathbb{E}[(\phi(T) - 1)\{L^2 - \psi_1(b)\}].$$

The subtraction of one is legitimate because the density's second derivative integrates to zero. Since $0 < \phi - 1 < (\sqrt{2} - 1)/2 < 1/4$ and

$$\psi_1(b) = \sum_{n \geq 0} (b + n)^{-2} \leq \zeta(2) < 2 \quad (b \geq 1),$$

we obtain $|C''(b)| < \frac{1}{4}(\mathbb{E}L^2 + \psi_1(b)) < 1$. The trigamma identity is classical [8]. The elementary upper bound $\zeta(2) < 2$ also follows by comparing its terms after the first with $\int_1^\infty x^{-2} dx$. $\square$

On an interval of length h , a function whose second derivative has absolute value at most one lies at most $h^2/8$ above the linear interpolation of its endpoints. This follows from the interpolation remainder or by comparison with the parabola $(x - x_0)(x_1 - x)/2$. Apply this on the two intervals between the three certified orders, with $h = 10^{-4}$. Together with (5.4), this bounds the maximum by the largest stored endpoint upper bound plus $1/(8 \cdot 10^8)$. Exact rational comparison yields

Theorem 5.2 (Certified universal constant). *The sharp universal Euler constant satisfies*

$$\boxed{1.1365611033 < C_* < 1.1365611046.} \quad (5.5)$$

Consequently 1.136562 is a valid rational universal constant for all $a, b > 0$ and $N \geq 1$. The simpler constant $57/50$ is also valid on the whole positive quadrant, not only on the critical triangle.

The interval in (5.5) is deliberately much shorter to state than the exact intermediate rational endpoints. Its width is 1.3×10^{-9} . It is a mathematical enclosure conditional only on the preceding ordinary proofs and finite integer checks, not a confidence interval from a numerical optimizer.

A separate 100-digit evaluation gives the diagnostics

$$\begin{aligned} b_* &\approx 1.3022165871012412095923717015551655, \\ C_* &\approx 1.1365611033395095609525863757799424, \\ C''(b_*) &\approx -0.0866705616486695948236951284473855. \end{aligned}$$

Only (5.4) and (5.5) are certified here. The longer decimals are retained as reproducible diagnostics, not silently promoted to rigorous digits.

5.3 A practical stopping rule

An exact or outward-rounded evaluation of E_N immediately gives

$$g_{a,b} \in [L - 1.136562 \cdot 2^{-N}, U] \quad (N \geq 1)$$

with the sharper lower endpoint $L - (9/8)2^{-N}$ whenever $N \geq 2$. For a target remainder $\varepsilon > 0$ and $N \geq 2$, it suffices to choose

$$N \geq \left\lceil \log_2 \frac{9}{8\varepsilon} \right\rceil.$$

Any uncertainty in forming the finite sum must be included separately in $U - L$. In particular, this is not a license to ignore cancellation in floating-point binomial sums. The exact-weight formula (5.1) and interval arithmetic provide a direct way to account for it.

6 An error generating function and all-order identities

6.1 Generating the complete error sequence

The remainder analysis gives more than a numerical estimate. Write

$$G_{a,b}(r) = \operatorname{Im} F_{a,b}(ir), \quad r > 0,$$

using the same analytic continuation as before. The positive imaginary axis never meets the principal cut.

Theorem 6.1 (Euler-error generating identity). *For $a \geq 0$, $b > 0$, and $0 < q < 2$, $q \neq 1$,*

$$\boxed{\sum_{N=1}^{\infty} e_N q^{N-1} = \frac{G_{a,b}(\sqrt{q/(2-q)}) / \sqrt{q(2-q)} - g_{a,b}}{1-q}}. \quad (6.1)$$

The series is absolutely convergent, locally uniformly for $|q| < 2$. The singularity displayed at $q = 1$ is removable.

Proof. For sufficiently small positive t , the binomial-transform generating function gives

$$\sum_{j \geq 0} d_j t^j = \frac{1}{1-t} \sum_{k \geq 0} A_k \left(\frac{-t}{1-t} \right)^k = \frac{G_{a,b}(r)}{r(1-t)}, \quad r = \sqrt{\frac{t}{1-t}}.$$

All series here converge absolutely for small t ; the coefficients A_k have at most polynomial growth, with a possible logarithmic factor. From $E_{N+1} - E_N = d_N/2^{N+1}$ we get

$$R_{N+1} = 2R_N + d_N, \quad R_0 = -g_{a,b}.$$

Consequently

$$\sum_{N \geq 0} R_N t^N = \frac{r G_{a,b}(r) - g_{a,b}}{1-2t}. \quad (6.2)$$

Set $t = q/2$, remove the $N = 0$ term and divide by q . This is (6.1) near zero.

The uniform remainder theorem proves absolute convergence of the left side throughout $|q| < 2$. On the real interval $0 < q < 2$, the numerator on the right is real analytic, vanishes at $q = 1$, and therefore has a removable quotient there. Real analytic continuation proves equality on the entire interval. Equivalently, one can complexify with the odd analytic function $[F(ir) - F(-ir)]/(2i)$; it agrees with $G(r)$ for real r . This avoids treating an imaginary-part operation as a holomorphic function of a complex variable. $\square$

6.2 Order-lowering polynomials

Define $P_n(\lambda) \in \mathbb{Q}[\lambda]$ by

$$\sum_{n \geq 0} P_n(\lambda) x^n = (1-x^2)^{-1/2} \exp\{\lambda \operatorname{arctanh} x\}. \quad (6.3)$$

They obey the finite recurrence

$$P_0 = 1, \quad P_1 = \lambda, \quad (n+1)P_{n+1} = \lambda P_n + n P_{n-1} \quad (n \geq 1). \quad (6.4)$$

Indeed $(1 - x^2)$ times the derivative of the generating function equals $(\lambda + x)$ times that function. In particular,

$$\begin{aligned} P_2 &= \frac{\lambda^2 + 1}{2}, & P_3 &= \frac{\lambda^3 + 5\lambda}{6}, \\ P_4 &= \frac{\lambda^4 + 14\lambda^2 + 9}{24}, & P_5 &= \frac{\lambda^5 + 30\lambda^3 + 89\lambda}{120}. \end{aligned}$$

Every nonzero coefficient is positive, and P_n has the parity of n .

Let $\mathcal{D} = z d/dz$. Termwise differentiation in the disk and analytic continuation give

$$\mathcal{D}^j F_{a,b}(z) = F_{a-j,b}(z). \quad (6.5)$$

When $a - j < 0$, the right side is defined first by the same disk series and then by this derivative continuation. The definition is consistent: choosing a larger positive starting order gives the same analytic function because the disk series agree. Set $g_{a-j,b} = \text{Im } F_{a-j,b}(i)$. The notation $P_n(\mathcal{D})g_{a,b}$ below means apply the operator to F first and then take its imaginary value at i ; it is not differentiation of a number with respect to z .

Theorem 6.2 (All-order binomial moments of the error). *For every integer $k \geq 0$, $a \geq 0$ and $b > 0$,*

$$\sum_{N \geq 1} \binom{N-1}{k} (E_N - g_{a,b}) = - \sum_{j=0}^{k+1} [\lambda^j] P_{k+1}(\lambda) g_{a-j,b}. \quad (6.6)$$

Every series on the left converges absolutely. Thus the following are exact identities, not fitted relations:

$$\sum_{N \geq 1} e_N = -g_{a-1,b}, \quad (6.7)$$

$$\sum_{N \geq 1} (N-1)e_N = -\frac{g_{a-2,b} + g_{a,b}}{2}, \quad (6.8)$$

$$\sum_{N \geq 1} \binom{N-1}{2} e_N = -\frac{g_{a-3,b} + 5g_{a-1,b}}{6}. \quad (6.9)$$

Proof. Put $q = 1 + x$ in (6.1). Then

$$\sqrt{\frac{q}{2-q}} = e^{\text{arctanh } x}, \quad \frac{1}{\sqrt{q(2-q)}} = (1 - x^2)^{-1/2}.$$

The Taylor expansion in the logarithm of the radius, together with (6.5), identifies the numerator as

$$\sum_{n \geq 0} \left\{ \sum_j [\lambda^j] P_n(\lambda) g_{a-j,b} \right\} x^n - g_{a,b}.$$

Dividing by $-x$ and comparing the coefficient of x^k proves the formula. The left-hand coefficient is obtained by expanding $(1 + x)^{N-1}$. The estimate $e_N < (9/8)2^{-N}$ for $N \geq 2$ justifies all termwise differentiations in a neighborhood of $q = 1$. $\square$

Corollary 6.3 (A certified tail for every moment). *If $M \geq 2$ and $M > 2k$, then the omitted tail of (6.6) satisfies*

$$0 < \sum_{N \geq M} \binom{N-1}{k} e_N < \frac{9}{8} \frac{2(M-k)}{M-2k} \binom{M-1}{k} 2^{-M}. \quad (6.10)$$

Proof. For $t_N = \binom{N-1}{k} 2^{-N}$, the ratio $t_{N+1}/t_N = N/[2(N-k)]$ is at most $M/[2(M-k)] < 1$ when $N \geq M$. Apply the geometric majorant and the remainder theorem. This also supplies explicit certified evaluation budgets for the new identities. $\square$

6.3 A Gaussian sign theorem at negative outer orders

Corollary 6.4 (Negativity through outer order minus one). *For every real $s \geq -1$ and every $b > 0$,*

$$g_{s,b} < 0. \quad (6.11)$$

Proof. In (6.7) take $a = s + 1 \geq 0$. Every error is strictly positive, so the right side $-g_{s,b}$ is strictly positive. $\square$

This corollary concerns analytic Gaussian values; it does not assert convergence of their original boundary series. Nor does it extend the Euler bound to negative outer orders. Higher moments provide additional strict linear inequalities among shifted-order values. For example,

$$g_{a-2,b} + g_{a,b} < 0, \quad g_{a-3,b} + 5g_{a-1,b} < 0 \quad (a \geq 0, b > 0).$$

A fixed sign cannot persist at all negative orders: the direct elementary evaluation below gives $g_{-3,1} = 1 + \pi/4 > 0$.

6.4 Explicit identities with rational harmonic Euler sums

For $(a, b) = (1, 1)$, the classical disk identity is

$$F_{1,1}(z) = \frac{1}{2} \log^2(1-z), \quad g_{1,1} = -\frac{\pi \log 2}{8}.$$

It follows by differentiating both sides and checking the zero value at the origin. The principal logarithm at $z = i$ is $\log(1-i) = \frac{1}{2} \log 2 - i\pi/4$. Let $E_N = E_N(1, 1)$ as defined in (1.2); each E_N is rational. Repeated application of $\mathcal{D}$ gives

$$\begin{aligned} g_{0,1} &= -\frac{\pi}{8} - \frac{\log 2}{4}, & g_{-1,1} &= -\frac{1}{2} - \frac{\pi}{8}, \\ g_{-2,1} &= -\frac{3}{4} + \frac{\log 2}{4}, & g_{-3,1} &= 1 + \frac{\pi}{4}. \end{aligned}$$

Substitution into the first three moment identities proves

$$\sum_{N \geq 1} \left(E_N + \frac{\pi \log 2}{8} \right) = \frac{\pi}{8} + \frac{\log 2}{4}, \quad (6.12)$$

$$\sum_{N \geq 1} (N-1) \left(E_N + \frac{\pi \log 2}{8} \right) = \frac{1}{4} + \frac{\pi(1 + \log 2)}{16}, \quad (6.13)$$

$$\sum_{N \geq 1} \binom{N-1}{2} \left(E_N + \frac{\pi \log 2}{8} \right) = \frac{1}{8} + \frac{\log 2}{6} + \frac{5\pi}{48}. \quad (6.14)$$

The parenthesized summands must be kept together. Separately summing the rational E_N and the added constant would split an absolutely convergent error series into divergent pieces.

There is also an exact axis derivative identity for every $b > 0$:

$$g_{-1,b} = -\frac{1}{2} \left\{ \beta(b) + \beta(b-1) + 2^{1-b} \eta(b-1) \right\}. \quad (6.15)$$

To verify it, apply $\mathcal{D}$ to $F_{0,b}(z) = z \operatorname{Li}_b(z)/(1-z)$, use $\mathcal{D} \operatorname{Li}_b = \operatorname{Li}_{b-1}$, and evaluate at i . The beta and eta terms at $b-1 \leq 0$ have their analytic meanings. Corollary 6.4 shows that the braces are strictly positive for every $b > 0$, an inequality that is not obvious from their individual continued-series notation.

7 Audit findings and proposed manuscript corrections

7.1 A targeted continuation, not a claim of a complete audit

This work inspected the current real-order, subcritical, Euler and research-programme sections and their immediate mathematical dependencies. It did not recheck all 375 pages of the source milestone. No false theorem in those inspected source sections is alleged here. In particular, the source correctly distinguished the Gaussian maximum from a universal Euler bound, and correctly restricted its proof of decreasing scaled errors to the appropriate positive-measure regime.

The principal proposed correction is now a *status change*: the universal real-order Euler question in the research programme can be replaced by Theorem 1.1. The parameter-dependent bounds in the old Euler section remain valid, but the new bound removes the need to use a diverging mass estimate as a universal budget. The critical constant should not be replaced by the larger global constant when working only in its smaller domain.

7.2 A concrete warning against an invalid shortcut

One might try to identify the universal Euler constant directly from the Gaussian maximum by claiming $R_N \leq R_1$. That shortcut is false, even for ordinary integer orders.

Proposition 7.1 (Scaled errors need not decrease). *At $(a, b) = (4, 1)$, $R_2 > R_1$.*

Proof. Since $A_1 = H_2/3^4 = 1/54$ and $E_2 = -A_1/4$, direct algebra gives

$$R_2 - R_1 = -2g_{4,1} - \frac{1}{54}.$$

The exact rational evaluation of (5.1) with $N = 48$ gives $-2E_{48}(4, 1) > 1/54$, as checked by `code/verify_exact.py`. Positivity of the remainder gives $g_{4,1} < E_{48}(4, 1)$ and proves the strict inequality. No numerical special-function value is used in this comparison. $\square$

For orientation only, $g_{4,1} \approx -0.01592246492806263368$. The proposition is a scope guard for future edits, not a correction of an assertion actually made by the manuscript. It explains why an independent later-truncation theorem was essential.

7.3 Precise editorial actions

Insert a section after the present real-order Euler bounds that states the positive divided-difference representation, the $9/8$ later-truncation bound, and the identification of C_* . The companion integration file uses a separate label prefix and points to this report for the finite certificates and complete proofs. Retain the earlier critical/subcritical theory because it supplies sharper domain-specific bounds and other geometric results.

In the research programme, replace the sentence leaving a universal real-order Euler bound open by the precise constant and the remaining optimization questions for fixed N . Add

the certified intervals (5.4)–(5.5) alongside, rather than in place of, the inherited diagnostic maximum. The old bracket $1 < b_* < 2$ was valid; the new bracket is a strengthening.

Do not promote the global integer-outer-order normalized-radius conjecture, the mixed-period S_6 candidate, or numerical period-independence questions. None is implied by the kernel argument. Likewise, do not interpret the positive difference quotient as a finite signed-moment representation in a parameter region where the source has proved that such a representation cannot exist.

Finally, distinguish $E_N - g$ from $2^N(E_N - g)$ in every monotonicity statement, and preserve the analytic/Abel convention at boundary points whose defining series diverges. These are mathematical hypotheses, not presentation details.

7.4 Independent evidence and its limits

The package includes three deliberately different verification layers. The dependency-free exact verifier proves polynomial identities, regenerates all rational Bernstein coefficients, checks integer power enclosures, and verifies finite inequalities. A SymPy replay independently derives the rational kernels and checks the generating polynomials and elementary identities. A separate numerical program compares finite Euler evaluations against elementary formulas and a depth-one integral representation.

The numerical computations use 100 decimal working digits. For $F_{1,1}$, 240 Euler terms give a discrepancy of about 1.2×10^{-73} from the elementary Gaussian value. Six error-moment comparisons have discrepancies ranging from about 1.2×10^{-73} to 7.8×10^{-64} . For $F_{1,2}$, the independent identity

$$F_{1,2}(z) = \int_0^z \frac{\text{Li}_2(t)}{1-t} dt$$

provides a path-quadrature comparison with a 200-term Euler evaluation; the observed discrepancy is about 2.8×10^{-61} . The differential identity follows directly from the disk series. These computations are useful independent diagnostics, but the quadrature itself is not interval-certified.

The exact proof of the main inequality depends on the stated analytic arguments and finite rational certificates. None of the programs formalizes measure theory, analytic continuation or differentiation under an integral in a proof assistant. A successful replay is also not evidence of transcendental independence.

8 Further research questions and a sharper kernel conjecture

The universal Euler constant is now determined, but the pointwise kernel bound is intentionally not optimized. This leaves several concrete problems whose formulations no longer depend on a broad numerical search for an unspecified constant.

8.1 The exact later-truncation kernel extrema

Define

$$M_N = \max_{0 \leq p, y \leq 1} K_N(p, y).$$

Continuity gives existence. Theorem 3.1 proves $M_N < 9/8$ for $N \geq 2$, while (4.1) gives $M_1 = (1 + \sqrt{2})/2$. Numerical optimization of the explicit rational function gives the following diagnostics:

N	p at a found maximizer	y at a found maximizer	Found value
2	0.852259	0.252703	1.118938686
3	0.692966	0.223626	1.097887069
4	0.583633	0.208496	1.087814658
8	0.361111	0.182102	1.072237961
16	0.206992	0.165668	1.063669537
32	0.112212	0.156110	1.058981067
128	0.030040	0.148143	1.055192383

These are floating-point optimization results, not certified global maxima.

Conjecture 8.1 (A monotone sequence of kernel maxima). *For every $N \geq 2$, K_N has a unique global maximizer in $(0, 1)^2$, and $M_{N+1} < M_N$.*

The monotonicity of the auxiliary envelope B_m proved earlier is not this conjecture: K_N and B_N are different functions. Likewise, R_N at a fixed order pair need not decrease, as Proposition 7.1 shows. A useful first target is an exact algebraic characterization and root-isolation certificate for the maximum of K_2 . Clearing the denominators of its two gradient equations produces a finite elimination problem; boundary comparisons must still be supplied.

8.2 An exponential limiting kernel

For fixed $c \geq 0$ and $0 \leq y \leq 1$, the scaling $p = c/N$ suggests

$$K_N(c/N, y) \longrightarrow H(c, y) := \frac{e^{-cy^2} - e^{-c}}{1 - y}. \quad (8.1)$$

At $y = 1$ the continuous limit is $H(c, 1) = 2ce^{-c}$.

Proposition 8.2 (Local scaling limit). *Convergence in (8.1) is uniform on every compact rectangle $0 \leq c \leq A$, $0 \leq y \leq 1$.*

Proof. For bounded $v \geq 0$ set $h_N(v) = f_N(v/N)$. The elementary binomial-to-exponential limit, and its first derivative, give $h_N \rightarrow e^{-v}$ in $C^1([0, A])$. For $y < 1$,

$$K_N(c/N, y) = \frac{h_N(cy^2) - h_N(c)}{1 - y} = -c(1 + y) \int_0^1 h'_N(cy^2 + uc(1 - y^2)) du.$$

Uniform convergence of the derivative proves the claim including $y = 1$. $\square$

For fixed $0 < y < 1$, H has a unique maximum in c at

$$c(y) = \frac{-2 \log y}{1 - y^2}, \quad H(c(y), y) = (1 + y)y^{2y^2/(1-y^2)}.$$

Both statements follow by a single differentiation. The endpoint limits of the last expression are one at $y = 0$ and $2/e$ at $y = 1$. At $y = 1/4$ its value exceeds one, since $(5/4)^{15} > 16$, so its maximum is attained in the interior. A numerical calculation finds $c \approx 3.9410662$, $y \approx 0.1453037$ and $H \approx 1.0538619304$.

Conjecture 8.3 (Limit of the optimal kernel constants). *The global maxima satisfy*

$$\lim_{N \rightarrow \infty} M_N = \max_{0 < y < 1} (1 + y)y^{2y^2/(1-y^2)}.$$

Moreover, the maximizers obey $Np_N \rightarrow c_\infty$ and $y_N \rightarrow y_\infty$, where (c_∞, y_∞) is the unique maximizer of H .

The proved compact scaling limit alone does not establish this assertion. A proof must exclude maximizing sequences escaping to $Np \rightarrow \infty$ or into problematic boundary regimes, and establish uniqueness for the limiting one-variable maximum. This makes the missing global step explicit.

8.3 Order-averaged optimality is a different problem

For fixed N , define $C_N = \sup_{a,b>0} \mathbb{E} K_N(X^2, Y)$. Then $C_N \leq M_N$, but equality is not automatic. Two gamma shape parameters do not allow arbitrary concentration at an independently selected interior point (p, y) with fixed rate one. Determine C_N , whether it is attained, and which parameter limits control it. The theorem of this report only needs $\sup_{N \geq 2} C_N < C_*$; it does not identify every C_N .

8.4 Further analytic and arithmetic directions

The error-moment identities suggest a structured route into negative outer orders. Determine, for each $b > 0$, the sign changes of $s \mapsto g_{s,b}$ below $s = -1$, and identify optimal linear inequalities from the positive coefficients of P_n . The explicit opposite signs at $g_{-1,1}$ and $g_{-3,1}$ show that a blanket extension of Gaussian negativity would be false. The moment identities also offer certified evaluations of selected negative-order combinations from rapidly convergent positive-order Euler errors.

Other acceleration schemes replace $(1 - v)^N$ by a different residual polynomial. Find conditions on such polynomials that control the divided difference uniformly. The positive-moment acceleration literature is relevant, but its theorems should not be applied by silently discarding the compensation factor $1 - Y$ in this problem. A useful goal is a class of admissible residual functions with explicit derivative or Bernstein certificates.

For other roots of unity, the remainder kernel generally becomes complex rather than positive. Determine whether a sector-valued or matrix-valued analogue gives equally uniform bounds. At higher depth, seek iterated positive differences rather than assuming that a depth-two proof automatically iterates without integrability losses.

The mixed Gaussian S_6 relation remains a separate arithmetic reduction problem. The present theorem can certify evaluations that support its investigation, but a vanishing numerical residual would still not prove that relation. The global normalized-radius conjecture also remains separate from the radial substitution in the error generating function.

Finally, the finite proof components are natural formalization targets: the Euler coefficient identity, the three polynomial identities, Bernstein basis conversion and exact integer-root intervals. Formalizing the probability kernel and analytic continuation would then connect those certificates to the global theorem. At present this is a proposed development, not an existing Lean certificate.

A Reproduction contract and certificate inventory

The package is self-contained for the exact finite replay. Python's standard library suffices for its core checker. The commands below are run from the package root:

```
python code/verify_exact.py
python -O code/verify_exact.py
python code/verify_sympy.py
python code/numerical_checks.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The last two Python programs require the dependencies documented in `requirements.txt`; SciPy is used only for optional numerical optimization. The exact checker uses explicit exceptions rather than Python assertions, so its checks remain active under `-O`.

A.1 Frozen data versus independent generation

The JSON polynomial file records every cell, degree and rational coefficient, not only a claimed minimum. The exact checker reconstructs the same data from the printed polynomial formulas. The symbolic checker instead reconstructs the rational kernels from f_N , clears their denominators, and independently converts the resulting polynomials to the Bernstein basis. Its separate generation route reduces the risk of an unnoticed copied coefficient error.

The axis JSON file records the proposed lower grid numerator for each rational power, the exact finite-sum interval and the reported global brackets. The checker verifies all defining integer inequalities before using these values. It also checks that the center axis interval lies above both neighbors and that the interpolation bound implies the claimed enclosure of C_* . The analytic premise $|C''| < 1$ is the written Lemma 5.1, not a numerical second-derivative estimate.

The expected finite inventory is:

Object checked	Count
Exact defining polynomial identities	3
Bernstein cells across all three certificates	82
Strictly positive rational Bernstein coefficients	1542
Independent Euler weight comparisons	528
Integer-certified rational power brackets	282
Axis function enclosures	3
Moment-polynomial generating coefficients, degrees 0–12	13

The corrupted-polynomial negative control is required to fail. The scaled-error monotonicity counterexample is checked as an exact finite rational inequality. The generator is not run by the read-only verifier; intentional regeneration is a separate command.

A.2 Numerical evidence

The numerical JSON is explicitly labeled nonrigorous. It records working precision, independently evaluated values, discrepancies, found kernel maximizers and the limiting-kernel optimization. The script fixes the optional optimizer’s random seed. The raw text reports are retained so that a later integration can compare a fresh replay with this delivery. These reports do not replace the source code or mathematical proofs.

A.3 Source and integration boundaries

The provenance file records the pinned source commit and the three source sections most directly affected. The integration directory contains a compact manuscript insertion and proposed editorial wording. Applying them requires review in the repository’s own build and reference environment. This delivery does not modify the remote repository, replace historical source archives, or regenerate the source manuscript’s inventory. Its standalone PDF is the compiled report associated with the supplied TeX sources.

B A short independent derivation of the envelope limit

The explicit maximum of J_m can also be used to see why the exponential problem is natural. If $y \in (0, 1)$ is fixed, then

$$B_m(y) = (1 + y) \left(\frac{1 - y^2}{1 - y^2 e^{2 \log y / (m-1)}} \right)^{m-1} \longrightarrow (1 + y) y^{2y^2 / (1-y^2)}.$$

This follows by taking logarithms and expanding the denominator to first order in $(m - 1)^{-1}$. It is consistent with the maximum of H computed in Section 8. However, the envelope convergence at fixed y still does not justify exchanging a global maximum and a limit without uniform endpoint control. This is another way to locate, rather than hide, the remaining step in Conjecture 8.3.

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